



# What You Should Learn in Python

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## PHASE 1 — Core Python

### Learn Properly

#### Topics

- variables
  - loops
  - functions
  - classes
  - OOP
  - exceptions
  - file handling
  - JSON
  - APIs
  - modules
  - virtual environments
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## PHASE 2 — Intermediate Python (VERY IMPORTANT)

#### Focus

- requests
- asyncio
- threading
- multiprocessing
- decorators
- generators

- logging
- testing
- type hints

These matter for senior engineering.

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## PHASE 3 — Backend + APIs

Learn:

- FastAPI
- REST APIs
- JWT auth
- async APIs
- Dockerized Python apps

This is HIGH ROI for your path.

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## PHASE 4 — Automation + Infra

Build:

- Kubernetes automation
  - GCP automation
  - CI/CD tooling
  - monitoring tools
  - log parsers
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## BEST UDEMY COURSE FOR YOU

### Start With

- 100 Days of Code: The Complete Python Pro Bootcamp by Angela Yu

But don't try to finish all 100 days sequentially.

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## YOUR CUSTOM PYTHON TRACK

**Focus ONLY on:**

| Priority | Topic      |
|----------|------------|
| High     | Functions  |
| High     | APIs       |
| High     | OOP        |
| High     | JSON       |
| High     | Async      |
| High     | Requests   |
| High     | FastAPI    |
| High     | Automation |

|        |            |
|--------|------------|
| Medium | Generators |
| Medium | Decorators |
| Low    | GUI        |
| Low    | Games      |

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## **BEST PRACTICAL LEARNING METHOD**

Don't study Python separately for months.

Instead:

### **Learn + Build Together**

Example:

#### **Week 1**

Learn:

- functions
- requests

Build:

- GCP VM inventory script
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#### **Week 2**

Learn:

- APIs
- FastAPI

Build:

- health-check API
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## Week 3

Learn:

- Kubernetes client

Build:

- pod monitoring tool
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# Python Projects PERFECT For Your Background

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## Project 1 — GKE Monitoring Tool

Build:

- pod health checker
  - namespace metrics
  - Slack alerts
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## Project 2 — Terraform Automation

Build:

- infra deployment helper
  - state validator
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## Project 3 — AI Inference API

Using:

- FastAPI
  - Docker
  - Kubernetes
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## Project 4 — Log Analysis Tool

Analyze:

- Kubernetes logs
  - GCP logs
  - latency spikes
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## SHOULD YOU LEARN GO?

For your path:

**YES — later.**

Because:

- Kubernetes ecosystem uses Go heavily
- cloud-native tooling uses Go
- operators/controllers often use Go

BUT:

✗ don't start there

Python gives MUCH faster ROI initially.

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# My Recommendation For YOU

## Step 1

Master practical Python deeply.

## Step 2

Use Python for:

- DSA
- APIs
- automation
- MLOps
- AI infra

## Step 3

After 6–12 months:  
learn Go basics.

That combination is VERY powerful:

- Python + Kubernetes + GCP + AI infra + some Go

That profile is extremely strong for modern platform engineering roles.